

Six-point homometry over fields: exact classification and counting of a classical family

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Abstract

We classify the image of a classical two-parameter pair of homometric six-point configurations under translation and inversion over fields of characteristic different from 2, 3, 11. Twelve explicit parameter symmetries account for every identification, including coincidences after projection. Centered moments reduce the classification to a cubic root-permutation problem. Over finite fields this gives exact counts; a single endpoint determines its partner except in characteristic 31, where an explicit counterexample is given. These are direct algebraic results. Total counts for six-subsets of prime cyclic groups additionally use an earlier computer-assisted completeness theorem and finite census. No historical novelty, external peer review, or proof-assistant certification is asserted.

1 The problem and the counted object

For a finite subset A of the additive group of a field K , define its directed autocorrelation by

$$c_A(h) = \#\{(x, y) \in A^2 : x - y = h\} \quad (h \in K).$$

Sets are *homometric* if their autocorrelations agree. This is a reconstruction problem from a difference multiset; it is relevant to crystallographic phase retrieval, sparse support recovery, and the cyclic beltway problem, as well as to musical interval content. For $K = \mathbb{F}_p$ with p an odd prime, it is equivalent to equality of cyclic interval-class vectors. For $K = \mathbb{F}_{p^r}$ with $r > 1$, the additive group is $(\mathbb{Z}/p\mathbb{Z})^r$, not the cyclic group $\mathbb{Z}/p^r\mathbb{Z}$.

Two sets are *rigidly equivalent* if $B = \epsilon A + t$, where $\epsilon \in \{1, -1\}$ and $t \in K$; write $[A]$ for a rigid class. General nonzero scalar multiplication is *not* identified. With $v = (a, b)^T \in K^2$, consider the classical configurations

$$\begin{aligned} X_v &= \{0, a, b - 2a, 2b - 2a, 2b, 3b - a\}, \\ Y_v &= \{0, a, b + 2a, 2b - a, 2b + a, 3b - a\}. \end{aligned}$$

Let Ω_K consist of parameters for which each list has six distinct entries, and put $\Phi(v) = \{[X_v], [Y_v]\}$. The endpoints of this unordered pair may be translated and inverted independently. We call the image of Φ the *Bloom image*; this is terminology for a classical family, not a priority assertion.

Postpischil–Gilbert give these formulas and their triangular projection geometry [1, §§3,7]. The issue here is the exact quotient after projection into a field. Formal symmetries alone do not exclude accidental identifications. The moment argument below supplies that missing step, without a distinct-distance assumption or a real ordering. Theorems A–C concern this specified image. They do not by themselves classify every homometric six-subset of every additive group.

2 Direct algebraic results

Throughout Theorems A–C, K has characteristic different from 2, 3, 11. The results are [PROVED], in-house, with a separate adversarial review recorded in the accompanying project note. The proofs below are algebraic; finite computations are supporting checks.

Theorem A (exact image classification). *Every $v \in \Omega_K$ gives two homometric, different rigid classes. Put*

$$R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad G = \{\pm I, \pm R, \pm R^2, \pm T, \pm RT, \pm R^2T\}.$$

Then G is a group of order twelve, isomorphic to $S_3 \times C_2$, which acts freely on Ω_K and preserves Φ . For $v, w \in \Omega_K$,

$$\Phi(v) = \Phi(w) \iff w \in Gv \iff (s(v), e(v)^2) = (s(w), e(w)^2),$$

where $s = a^2 - ab + b^2$ and $e = ab(a - b)$. Thus every unordered pair has exactly twelve parameter representatives.

Theorem B (finite-field enumeration). *For $K = \mathbb{F}_q$ in the stated characteristic, the number of distinct unordered pairs in the Bloom image is*

$$B(q) = \begin{cases} (q-1)(q-5)/12, & \text{char } K = 5, \\ (q-1)(q-7)/12, & \text{char } K = 7, \\ (q-1)(q-11)/12, & \text{char } K \geq 13. \end{cases}$$

In particular $B(p) = (p-1)(p-11)/12$ for every prime $p \geq 13$. The displayed configurations give no six-element pair in $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{11}$.

The last assertion is a support observation, not an extension of the classification proof to characteristic 11. Extension fields of characteristics 2, 3, 11 are outside Theorems A–B.

Theorem C (unique partner within the image). *If $\text{char } K \neq 31$ as well, two distinct pairs in the Bloom image cannot share a rigid class. Consequently the graph with rigid classes as vertices and Bloom pairs as edges is a disjoint union of edges.*

Theorem C permits additional homometric partners outside the Bloom image. Characteristic 31 is a genuine exception: Section 6 gives two different Bloom pairs sharing one endpoint. Theorems A–B continue to hold there.

For example, $B(13) = 2$, $B(31) = 50$, and $B(1009) = 83\,832$. The counts in the noncyclic additive fields $\mathbb{F}_{25}, \mathbb{F}_{49}, \mathbb{F}_{169}$ are respectively 40, 168, 2212. These field counts retain translation and inversion as their sole equivalences. A quotient by all field scalars or by all additive-group automorphisms would be a different problem.

3 Differences, collisions, and the twelve symmetries

The fifteen differences of either configuration, taken up to sign and with multiplicity, are the following common list:

$$a, b - 2a, 2b - 2a, 2b, 3b - a, b - 3a, 2b - 3a, 2b - a, \\ 3b - 2a, b, b + 2a, 2b + a, 2a, b - a, b + a.$$

Replace each entry ℓ by $\ell, -\ell$ and include the six diagonal zero differences. This gives equal directed autocorrelations whenever both supports have size six. It is an integer coefficient identity; repeated nonzero differences cause no difficulty. Reduction into an additive group preserves the identity for the labeled lists, but support collisions must first be excluded to interpret them as six-element sets.

As 2 is invertible, collisions occur exactly on the union of the following twelve kernels:

$$a, b, b - a, b + a, b - 2a, b + 2a, b - 3a, \\ 2b - a, 2b + a, 2b - 3a, 3b - a, 3b - 2a. \quad (1)$$

Some kernels coincide in characteristics 5 and 7; the union remains exact. In particular, for every admissible parameter,

$$e = ab(a - b) \neq 0, \quad d = (a + b)(2a - b)(a - 2b) \neq 0.$$

The relations $R^3 = T^2 = I$, $TRT = R^{-1}$, and central $-I$ show that $G \cong S_3 \times C_2$. Its twelve matrices are distinct in characteristic different from 2, 3. Direct substitution gives the following identities of sets:

M	X_{Mv}	Y_{Mv}
I	X_v	Y_v
R	$X_v + 2a - 2b$	$Y_v + a - 3b$
R^2	$X_v - 2b$	$Y_v - 2a - b$
T	$Y_v + a - 2b$	$X_v + 2a - b$
RT	$Y_v - a$	$X_v - a$
R^2T	$Y_v - a - 2b$	$X_v + a - 3b$

For $-M$, negate the sets and translations on the right. Hence the whole group preserves admissibility and Φ .

To prove freeness, note that $\det(R - I) = \det(R^2 - I) = 3$ and $\det(R + I) = \det(R^2 + I) = 1$. Thus $\pm R, \pm R^2$, and $-I$ have only the zero fixed vector. The fixed lines of $T, -T, RT, -RT, R^2T, -R^2T$ are, respectively,

$$b = a, \quad b = -a, \quad a = 0, \quad a = 2b, \quad b = 0, \quad b = 2a.$$

Each is excluded by (1); zero is excluded as well. Every orbit therefore has twelve parameters. This still leaves open whether distinct orbits can become the same pair after projection.

4 Integral centered moments recover the orbit invariants

For a six-element set $A \subset K$, set

$$\mu_A = \frac{1}{6} \sum_{x \in A} x, \quad m_j(A) = \sum_{x \in A} (3(x - \mu_A))^j.$$

Translation preserves each m_j and inversion multiplies it by $(-1)^j$. Thus m_2, m_3^2, m_6 are invariants of a rigid class; their endpoint sums are invariants of an unordered pair.

All identities needed here can be checked over $\mathbb{Z}$. Indeed $3(x - \mu_A) = 3x - \frac{1}{2} \sum A$, while $\sum X_v = -4a + 8b$ and $\sum Y_v = 2a + 8b$. The scaled centered coordinates are the integral linear forms

$$\begin{aligned} X &: (2a - 4b, 5a - 4b, -4a - b, -4a + 2b, 2a + 2b, -a + 5b), \\ Y &: (-a - 4b, 2a - 4b, 5a - b, -4a + 2b, 2a + 2b, -4a + 5b). \end{aligned}$$

Raising these six forms to the stated powers and summing gives

$$d^2 + 27e^2 = 4s^3, \tag{2}$$

$$m_2(X_v) = m_2(Y_v) = 66s, \tag{3}$$

$$m_3(X_v) = 6d - 243e, \quad m_3(Y_v) = 6d + 243e, \tag{4}$$

$$m_6(X_v) = 23946s^3 + 6399e^2 - 4860de, \tag{5}$$

$$m_6(Y_v) = 23946s^3 + 6399e^2 + 4860de. \tag{6}$$

For an explicit expansion check, the coefficients of the summed sixth moments in the monomial order $a^6, a^5b, \dots, b^6$ are

$$(47892, -143676, 300150, -360840, 300150, -143676, 47892),$$

the same as those of $47892s^3 + 12798e^2$.

Define invariant endpoint sums

$$C_3 = m_3(X_v)^2 + m_3(Y_v)^2 - 288s^3 = 116154e^2,$$

$$C_6 = m_6(X_v) + m_6(Y_v) - 47892s^3 = 12798e^2.$$

The integer relations

$$116154 = 162 \cdot 717, \quad 12798 = 162 \cdot 79, \quad 118 \cdot 79 - 13 \cdot 717 = 1$$

yield the exact elimination identity

$$\boxed{118C_6 - 13C_3 = 162e^2.} \tag{7}$$

If $\Phi(v) = \Phi(w)$, equation (3) first recovers the common s , since 66 is invertible in the stated characteristic. The endpoint sums then recover C_3, C_6 , and (7) recovers e^2 , since 162 is invertible. Using both moments avoids exclusions at characteristics 79 and 239, where one individual coefficient vanishes. There is no division by s : isotropic parameters $s = 0$ are included.

5 Cubic root permutations and finite-field counting

Suppose $s(v) = s(w)$ and $e(v)^2 = e(w)^2$. As the characteristic is odd, $e(w) = e(v)$ or $e(w) = -e(v)$. In the latter case replace w by $-w$, an element of G , so that the values of e agree. The identity

$$(z - a)(z + b)(z - b + a) = z^3 - sz - e \quad (8)$$

then gives the same multiset of roots for the two ordered triples $(a, -b, b - a)$. This includes roots with multiplicity: uniqueness of factorization over a field is sufficient.

The map R cyclically permutes this ordered triple; $-T$ transposes its first two entries and fixes the third. They generate all six permutations. The first two entries recover a, b , so the aligned w is in the group generated by $R, -T$. Restoring any negation places the original w in Gv . Conversely the explicit symmetries preserve Φ and its recovered invariants. Thus equality of the invariants, equality of pairs, and membership in one G -orbit are equivalent.

The two endpoints cannot collapse to one rigid class. From (4),

$$m_3(X_v)^2 - m_3(Y_v)^2 = -5832de \neq 0.$$

The factors d, e are nonzero on Ω_K , and 5832 has only 2, 3 as prime factors. Rigid equivalence would preserve the squared third moment, a contradiction. This proves Theorem A in full, including the absence of accidental projected identifications.

For Theorem B, the primitive normals of the collision lines are

$$(1, 0), (0, 1), (-1, 1), (1, 1), (-2, 1), (2, 1), (-3, 1), \\ (-1, 2), (1, 2), (-3, 2), (-1, 3), (-2, 3).$$

The nonzero absolute pairwise determinants are exactly $\{1, 2, 3, 4, 5, 7, 8\}$. Thus the twelve lines are distinct in characteristic at least 13. Their projective slopes b/a are

$$\infty, 0, 1, -1, 2, -2, 3, 1/2, -1/2, 3/2, 1/3, 2/3.$$

In characteristic 5 these reduce to all six directions of $\mathbb{P}^1(\mathbb{F}_5)$; in characteristic 7, to all eight directions of $\mathbb{P}^1(\mathbb{F}_7)$. They remain six or eight distinct lines over an extension field, since distinct base-field slopes remain distinct.

Let d_p be this number of lines over $K = \mathbb{F}_q$. Each contains $q - 1$ nonzero vectors, and distinct lines meet only at zero. Therefore

$$|\Omega_K| = q^2 - 1 - d_p(q - 1) = (q - 1)(q + 1 - d_p).$$

With $d_5 = 6$, $d_7 = 8$, and $d_p = 12$ for $p \geq 13$, divide by twelve using Theorem A to obtain the three formulas in Theorem B. For $\mathbb{F}_{11}$ the slopes cover all twelve projective directions, so there is no admissible parameter. The fields $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5$ have fewer than six elements; $\mathbb{F}_7$ is already covered by the direction argument.

6 A single endpoint and the characteristic-31 obstruction

A rigid class occurring at either endpoint first recovers s from m_2 . For an endpoint $A = X_v$, equations (2)–(6) give

$$\begin{pmatrix} m_3(A)^2 - 144s^3 \\ m_6(A) - 23946s^3 \end{pmatrix} = \begin{pmatrix} 58077 & -2916 \\ 6399 & -4860 \end{pmatrix} \begin{pmatrix} e^2 \\ de \end{pmatrix}. \quad (9)$$

For $A = Y_v$ the same matrix applies with second unknown $-de$. Its determinant is

$$58077(-4860) - (-2916)6399 = -263594736 = -2^4 3^{12} 31.$$

Outside characteristics 2, 3, 11, 31, the endpoint therefore determines s, e^2 , regardless of which side it occupies. Two Bloom pairs sharing an endpoint have the same invariants and hence, by Theorem A, are the same pair. This proves Theorem C. It again requires no division by s or any data-dependent value.

The exceptional characteristic is necessary. In $\mathbb{F}_{31}$ take $v = (2, 12)$ and $w = (8, 17)$, and set

$$\begin{aligned} A &= \{0, 1, 3, 8, 12, 18\}, \\ B &= \{0, 1, 3, 10, 14, 26\}, \\ C &= \{0, 1, 4, 10, 12, 17\}. \end{aligned}$$

The raw configurations are

$$\begin{aligned} X_v &= \{0, 2, 3, 8, 20, 24\}, & Y_v &= \{0, 2, 3, 16, 22, 26\}, \\ X_w &= A, & Y_w &= \{0, 2, 8, 11, 12, 26\}. \end{aligned}$$

Their exact transports are

$$A = 3 - Y_v = X_w, \quad B = 3 - X_v, \quad C = 12 - Y_w.$$

Both parameters are admissible, so $\{[A], [B]\}$ and $\{[A], [C]\}$ are Bloom pairs. All three sets have each cyclic interval $1, \dots, 15$ exactly once. Their sorted multisets of successive circular gaps are, respectively,

$$(1, 2, 4, 5, 6, 13), \quad (1, 2, 4, 5, 7, 12), \quad (1, 2, 3, 5, 6, 14).$$

Translation rotates a circular gap sequence and inversion reverses it; the different gap multisets exclude rigid equivalence. Thus the two pairs genuinely differ and share exactly the stated class.

Moreover, $s(v) = s(w) = 0$ but $e(v)^2 = 2$ and $e(w)^2 = 8$ in $\mathbb{F}_{31}$. This exhibits failure of endpoint recovery, not merely a singular matrix in one method of proof. Pair classification and the count $B(31) = 50$ remain valid. The existence of this three-class subfamily alone does not establish the complete size-five family asserted by the finite census in the next section.

7 Total counts in prime cyclic groups: inherited dependencies

The direct algebra above counts the Bloom image. To identify it with *all* nontrivial six-point homometry in $\mathbb{Z}/p\mathbb{Z}$ at large primes, we use a separate completeness input from the Homometry Program:

Input G and its large-prime corollary. The earlier general six-point generation theorem is [PROVED], in-house, *computer-assisted*. Its large-prime corollary states that every cyclic six-point pair of different rigid classes is an admissible modular Bloom pair when every prime divisor of the modulus exceeds 131. This uses the exact line-classification solver obligation and the bounded universal-presentation/torsion certificate. The torsion order is at most 135; the least prime above 131 is 137, so the torsion image vanishes. Integer lifts then reduce the free image to the line classification. Those computational obligations are inherited, not proved or replaced by the present moment argument.

Corollary D (large-prime totals, assuming Input G). *For every prime $p > 131$, the number of unordered nontrivial six-point pair edges, and the number of maximal nontrivial six-point families in $\mathbb{Z}/p\mathbb{Z}$, are both $(p-1)(p-11)/12$. Each family has exactly two rigid classes.*

Proof. Input G identifies every pair with the Bloom image. Theorem B counts its edges, and Theorem C makes the graph a disjoint union of edges, since these primes differ from 31. $\square$

Corollary E (all prime totals, with finite census input). *In addition to Input G, assume the earlier complete two-method six-set census through modulus 135. For primes $p \geq 13$, put $B(p) = (p-1)(p-11)/12$ and let $[p=r]$ be 1 if $p=r$ and 0 otherwise. The total numbers of pair edges and maximal nontrivial families are*

$$\begin{aligned} P_6(p) &= B(p) + 8[p=17] + 9[p=19] + 11[p=23] + 20[p=31], \\ F_6(p) &= B(p) + 8[p=17] + 9[p=19] + 11[p=23] + 11[p=31]. \end{aligned}$$

Every family has size two except one size-five family in $\mathbb{Z}/31\mathbb{Z}$. Primes below 13 have no nontrivial six-set family.

Proof. Corollary D covers primes above 131. For the remaining primes the complete census gives extra edge counts 8, 9, 11, 20 at 17, 19, 23, 31, and zero elsewhere. At 31 there are sixty size-two families and one size-five family, hence 61 families and $60 + \binom{5}{2} = 70$ edges; subtracting $B(31) = 50$ gives the two different corrections. At 17, 19, 23 all families have size two. Primes below 7 support no six-subsets, and the complete checks at 7, 11 give no nontrivial families. $\square$

Corollaries D–E are [PROVED], in-house, *computer-assisted* under these recorded inputs. The small-prime replay checks canonical classes, autocorrelations, disjoint membership, family sizes and source hashes; its completeness is inherited from the original enumeration. It is not a new exhaustive census or a pure algebraic classification of the small-prime additional pairs.

8 Verification, provenance, and remaining scope

The mathematical source is `notes/2026-09-30-six-prime-theorem.md`; its separate attack is `notes/2026-09-30-six-prime-review.md`. An auxiliary coefficient-symmetry derivation and an independent polynomial kernel check all displayed identities, collision determinants, and the single-endpoint determinant. The proofs in Theorems A–C require only those written identities and elementary field algebra.

Two independent implementations provide [COMPUTED] controls. The direct point enumerator canonicalizes each endpoint under translation and inversion and retains every parameter fiber, without using the formula or symmetry group. The invariant implementation partitions parameters by (s, e^2) without constructing point classes. The *entire partitions* agree at every prime $13 \leq p \leq 251$ and at 1009, and at the additive fields $\mathbb{F}_{25}, \mathbb{F}_{49}, \mathbb{F}_{169}$. These checks support the algebraic proof and do not supply general completeness. Tests cover isotropic parameters, the coefficient exceptions 79, 239, and the explicit characteristic-31 shared endpoint.

The retained complete six-set census has its original two independent methods. Replaying its small-prime payloads validates the corrections in Corollary E while retaining the original completeness dependency. Full source and evidence hashes, native compiler diagnostics, exact export commands and page inspection are saved alongside this PDF. Useful reproduction commands in the pinned project environment are:

```
.venv/bin/python tests/run_tests.py
.venv/bin/python tests/test_six_prime_bloom.py
.venv/bin/python tests/test_six_bloom_invariants.py
.venv/bin/python src/six_bloom_invariants.py
```

The large-prime input is in the earlier working paper’s section “Integer shadows and the large-prime boundary”; `notes/2026-09-30-six-paper.tex`. Its proof and certificate obligations are not replaced by this standalone paper.

Prior art and limits. The family and its triangular geometry are classical [1]. Ranieri et al. display the same six-point matrices in the collision-free real-line setting [2, §IV.A]; that scope is distinct from modular coincidences. The earlier small-prime family totals 2, 16, 21, 33 at 13, 17, 19, 23 appear in Jedrzejewski–Johnson [3, §3]; family counts and pair-edge counts differ when a family has more than two members. The focused primary-source literature log is `notes/2026-09-30-six-prime-literature.md`. Database access failures and unread historical sources remain recorded; no novelty is claimed. Composite cyclic moduli, a complete classification over noncyclic additive fields, and extension fields of characteristics 2, 3, 11 remain outside this paper. External specialist review is still needed before public dissemination; no submission or external publication has occurred.

References

- [1] E. Postpischil and P. Gilbert, There Are No New Homometric Golomb Ruler Pairs with 12 Marks or Less, *Experimental Mathematics* **3**(2) (1994), 147–152. doi:10.1080/10586458.1994.10504286. Primary sections used: 3 and 7.
- [2] J. Ranieri, A. Chebira, Y. M. Lu and M. Vetterli, *Phase Retrieval for Sparse Signals: Uniqueness Conditions*, arXiv:1308.3058v2 (2013), §IV.A.
- [3] F. Jedrzejewski and T. Johnson, *The Structure of Z-Related Sets*, arXiv:1304.6608v1 (2013), §3; MCM 2013, LNCS 7937, 128–137.